AMATH 362: Oral Test Takehome, Dec. 14,15
Prepare up to 5 slides for presentation

Consider the simplest SHO with exponential damping

$$x'' + \exp(-\epsilon v)v + x = 0$$

where

$$v = \frac{dx}{dt}$$

and $\epsilon \ll 1$.

i) Derive the energy equation and discuss how it compares/contrasts with the damped SHO.

ii) Compare the exponential damping and damping that is proportional to an integer power of v. Use techniques from the course.

iii) Consider the exponential damping and a version with linearized damping (the ODE is still nonlinear),

$$x'' + (1 - \epsilon v)v + x = 0$$

Let $v(0) = 0$ and vary ϵ and $x(0)$ to discuss how similar or different the two systems are. Are there any advantages to using the simplified system?